

Problem 10.26

A fund earns interest at a constant force of interest of δ . Kathy invested \$10,000 in the fund. Twenty-two years later, Kathy has \$30,042. Calculate δ .

Solution

Under a constant force of interest δ , an initial investment P accumulates for t years according to

$$A(t) = Pe^{\delta t}.$$

Here,

$$P = 10,000, \quad A(22) = 30,042.$$

Therefore,

$$30,042 = 10,000e^{22\delta}.$$

Dividing by 10,000 and taking natural logarithms gives

$$\begin{aligned} e^{22\delta} &= \frac{30,042}{10,000}, \\ 22\delta &= \ln\left(\frac{30,042}{10,000}\right). \end{aligned}$$

Hence,

$$\begin{aligned} \delta &= \frac{1}{22} \ln\left(\frac{30,042}{10,000}\right) \\ &\approx 0.050000514072. \end{aligned}$$

Thus, the constant force of interest is approximately

$$\boxed{\delta \approx 5.0000514\%}.$$